

EXERCISE 8.1

- In $\triangle ABC$, right-angled at B, $AB = 24$ cm, $BC = 7$ cm. Determine :
 - $\sin A$, $\cos A$
 - $\sin C$, $\cos C$
- In Fig. 8.13, find $\tan P - \cot R$.
- If $\sin A = \frac{3}{4}$, calculate $\cos A$ and $\tan A$.
- Given $15 \cot A = 8$, find $\sin A$ and $\sec A$.
- Given $\sec \theta = \frac{13}{12}$, calculate all other trigonometric ratios.
- If $\angle A$ and $\angle B$ are acute angles such that $\cos A = \cos B$, then show that $\angle A = \angle B$.
- If $\cot \theta = \frac{7}{8}$, evaluate : (i) $\frac{(1 + \sin \theta)(1 - \sin \theta)}{(1 + \cos \theta)(1 - \cos \theta)}$, (ii) $\cot^2 \theta$
- If $3 \cot A = 4$, check whether $\frac{1 - \tan^2 A}{1 + \tan^2 A} = \cos^2 A - \sin^2 A$ or not.
- In triangle ABC, right-angled at B, if $\tan A = \frac{1}{\sqrt{3}}$, find the value of:
 - $\sin A \cos C + \cos A \sin C$
 - $\cos A \cos C - \sin A \sin C$
- In $\triangle PQR$, right-angled at Q, $PR + QR = 25$ cm and $PQ = 5$ cm. Determine the values of $\sin P$, $\cos P$ and $\tan P$.
- State whether the following are true or false. Justify your answer.
 - The value of $\tan A$ is always less than 1.
 - $\sec A = \frac{12}{5}$ for some value of angle A.
 - $\cos A$ is the abbreviation used for the cosecant of angle A.
 - $\cot A$ is the product of $\cot$ and A.
 - $\sin \theta = \frac{4}{3}$ for some angle θ .

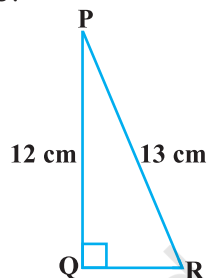


Fig. 8.13

8.3 Trigonometric Ratios of Some Specific Angles

From geometry, you are already familiar with the construction of angles of 30° , 45° , 60° and 90° . In this section, we will find the values of the trigonometric ratios for these angles and, of course, for 0° .

Trigonometric Ratios of 45°

In $\triangle ABC$, right-angled at B, if one angle is 45° , then the other angle is also 45° , i.e., $\angle A = \angle C = 45^\circ$ (see Fig. 8.14).

So, $BC = AB$ (Why?)

Now, Suppose $BC = AB = a$.

Then by Pythagoras Theorem, $AC^2 = AB^2 + BC^2 = a^2 + a^2 = 2a^2$,

and, therefore, $AC = a\sqrt{2}$.

Using the definitions of the trigonometric ratios, we have :

$$\sin 45^\circ = \frac{\text{side opposite to angle } 45^\circ}{\text{hypotenuse}} = \frac{BC}{AC} = \frac{a}{a\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\cos 45^\circ = \frac{\text{side adjacent to angle } 45^\circ}{\text{hypotenuse}} = \frac{AB}{AC} = \frac{a}{a\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\tan 45^\circ = \frac{\text{side opposite to angle } 45^\circ}{\text{side adjacent to angle } 45^\circ} = \frac{BC}{AB} = \frac{a}{a} = 1$$

$$\text{Also, } \operatorname{cosec} 45^\circ = \frac{1}{\sin 45^\circ} = \sqrt{2}, \sec 45^\circ = \frac{1}{\cos 45^\circ} = \sqrt{2}, \cot 45^\circ = \frac{1}{\tan 45^\circ} = 1.$$

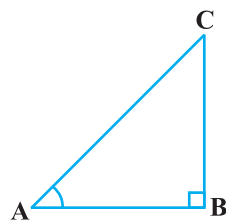


Fig. 8.14

Trigonometric Ratios of 30° and 60°

Let us now calculate the trigonometric ratios of 30° and 60° . Consider an equilateral triangle ABC. Since each angle in an equilateral triangle is 60° , therefore, $\angle A = \angle B = \angle C = 60^\circ$.

Draw the perpendicular AD from A to the side BC (see Fig. 8.15).

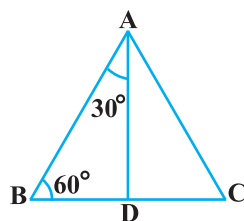


Fig. 8.15

Now $\triangle ABD \cong \triangle ACD$ (Why?)

Therefore, $BD = DC$

and $\angle BAD = \angle CAD$ (CPCT)

Now observe that:

$\triangle ABD$ is a right triangle, right-angled at D with $\angle BAD = 30^\circ$ and $\angle ABD = 60^\circ$ (see Fig. 8.15).

As you know, for finding the trigonometric ratios, we need to know the lengths of the sides of the triangle. So, let us suppose that $AB = 2a$.

Then,
$$BD = \frac{1}{2}BC = a$$

and
$$AD^2 = AB^2 - BD^2 = (2a)^2 - (a)^2 = 3a^2,$$

Therefore,
$$AD = a\sqrt{3}$$

Now, we have :

$$\sin 30^\circ = \frac{BD}{AB} = \frac{a}{2a} = \frac{1}{2}, \quad \cos 30^\circ = \frac{AD}{AB} = \frac{a\sqrt{3}}{2a} = \frac{\sqrt{3}}{2}$$

$$\tan 30^\circ = \frac{BD}{AD} = \frac{a}{a\sqrt{3}} = \frac{1}{\sqrt{3}}.$$

Also,
$$\operatorname{cosec} 30^\circ = \frac{1}{\sin 30^\circ} = 2, \quad \sec 30^\circ = \frac{1}{\cos 30^\circ} = \frac{2}{\sqrt{3}}$$

$$\cot 30^\circ = \frac{1}{\tan 30^\circ} = \sqrt{3}.$$

Similarly,

$$\sin 60^\circ = \frac{AD}{AB} = \frac{a\sqrt{3}}{2a} = \frac{\sqrt{3}}{2}, \quad \cos 60^\circ = \frac{1}{2}, \quad \tan 60^\circ = \sqrt{3},$$

$$\operatorname{cosec} 60^\circ = \frac{2}{\sqrt{3}}, \quad \sec 60^\circ = 2 \text{ and } \cot 60^\circ = \frac{1}{\sqrt{3}}.$$

Trigonometric Ratios of 0° and 90°

Let us see what happens to the trigonometric ratios of angle A , if it is made smaller and smaller in the right triangle ABC (see Fig. 8.16), till it becomes zero. As $\angle A$ gets smaller and smaller, the length of the side BC decreases. The point C gets closer to point B , and finally when $\angle A$ becomes very close to 0° , AC becomes almost the same as AB (see Fig. 8.17).

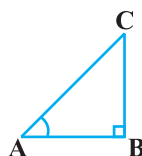


Fig. 8.16

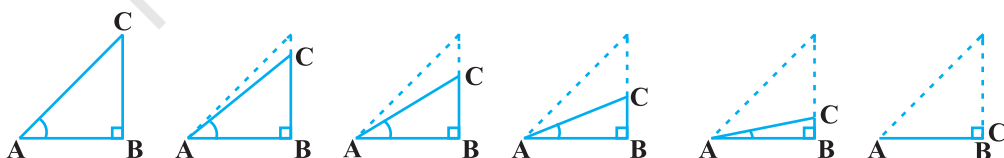


Fig. 8.17

When $\angle A$ is very close to 0° , BC gets very close to 0 and so the value of $\sin A = \frac{BC}{AC}$ is very close to 0. Also, when $\angle A$ is very close to 0° , AC is nearly the same as AB and so the value of $\cos A = \frac{AB}{AC}$ is very close to 1.

This helps us to see how we can define the values of $\sin A$ and $\cos A$ when $A = 0^\circ$. We define : **$\sin 0^\circ = 0$ and $\cos 0^\circ = 1$** .

Using these, we have :

$$\tan 0^\circ = \frac{\sin 0^\circ}{\cos 0^\circ} = 0, \cot 0^\circ = \frac{1}{\tan 0^\circ}, \text{ which is not defined. (Why?)}$$

$$\sec 0^\circ = \frac{1}{\cos 0^\circ} = 1 \text{ and } \operatorname{cosec} 0^\circ = \frac{1}{\sin 0^\circ}, \text{ which is again not defined. (Why?)}$$

Now, let us see what happens to the trigonometric ratios of $\angle A$, when it is made larger and larger in $\triangle ABC$ till it becomes 90° . As $\angle A$ gets larger and larger, $\angle C$ gets smaller and smaller. Therefore, as in the case above, the length of the side AB goes on decreasing. The point A gets closer to point B . Finally when $\angle A$ is very close to 90° , $\angle C$ becomes very close to 0° and the side AC almost coincides with side BC (see Fig. 8.18).

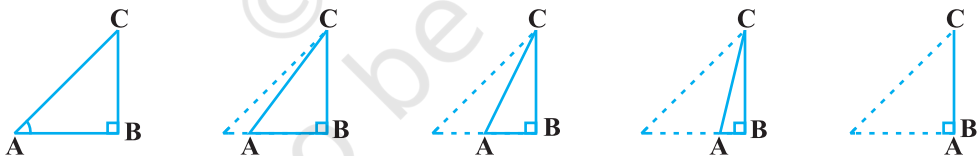


Fig. 8.18

When $\angle C$ is very close to 0° , $\angle A$ is very close to 90° , side AC is nearly the same as side BC , and so $\sin A$ is very close to 1. Also when $\angle A$ is very close to 90° , $\angle C$ is very close to 0° , and the side AB is nearly zero, so $\cos A$ is very close to 0.

So, we define : **$\sin 90^\circ = 1$ and $\cos 90^\circ = 0$** .

Now, why don't you find the other trigonometric ratios of 90° ?

We shall now give the values of all the trigonometric ratios of 0° , 30° , 45° , 60° and 90° in Table 8.1, for ready reference.

EXERCISE 8.2

1. Evaluate the following :

(i) $\sin 60^\circ \cos 30^\circ + \sin 30^\circ \cos 60^\circ$

(ii) $2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$

(iii) $\frac{\cos 45^\circ}{\sec 30^\circ + \operatorname{cosec} 30^\circ}$

(iv) $\frac{\sin 30^\circ + \tan 45^\circ - \operatorname{cosec} 60^\circ}{\sec 30^\circ + \cos 60^\circ + \cot 45^\circ}$

(v) $\frac{5 \cos^2 60^\circ + 4 \sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$

2. Choose the correct option and justify your choice :

(i) $\frac{2 \tan 30^\circ}{1 + \tan^2 30^\circ} =$

(A) $\sin 60^\circ$

(B) $\cos 60^\circ$

(C) $\tan 60^\circ$

(D) $\sin 30^\circ$

(ii) $\frac{1 - \tan^2 45^\circ}{1 + \tan^2 45^\circ} =$

(A) $\tan 90^\circ$

(B) 1

(C) $\sin 45^\circ$

(D) 0

(iii) $\sin 2A = 2 \sin A$ is true when $A =$

(A) 0°

(B) 30°

(C) 45°

(D) 60°

(iv) $\frac{2 \tan 30^\circ}{1 - \tan^2 30^\circ} =$

(A) $\cos 60^\circ$

(B) $\sin 60^\circ$

(C) $\tan 60^\circ$

(D) $\sin 30^\circ$

3. If $\tan (A + B) = \sqrt{3}$ and $\tan (A - B) = \frac{1}{\sqrt{3}}$; $0^\circ < A + B \leq 90^\circ$; $A > B$, find A and B .

4. State whether the following are true or false. Justify your answer.

(i) $\sin (A + B) = \sin A + \sin B$.

(ii) The value of $\sin \theta$ increases as θ increases.

(iii) The value of $\cos \theta$ increases as θ increases.

(iv) $\sin \theta = \cos \theta$ for all values of θ .

(v) $\cot A$ is not defined for $A = 0^\circ$.

$$= \frac{-1}{\tan \theta - \sec \theta} = \frac{1}{\sec \theta - \tan \theta},$$

which is the RHS of the identity, we are required to prove.

EXERCISE 8.3

- Express the trigonometric ratios $\sin A$, $\sec A$ and $\tan A$ in terms of $\cot A$.
- Write all the other trigonometric ratios of $\angle A$ in terms of $\sec A$.
- Choose the correct option. Justify your choice.
 - $9 \sec^2 A - 9 \tan^2 A =$
 (A) 1 (B) 9 (C) 8 (D) 0
 - $(1 + \tan \theta + \sec \theta)(1 + \cot \theta - \operatorname{cosec} \theta) =$
 (A) 0 (B) 1 (C) 2 (D) -1
 - $(\sec A + \tan A)(1 - \sin A) =$
 (A) $\sec A$ (B) $\sin A$ (C) $\operatorname{cosec} A$ (D) $\cos A$
 - $\frac{1 + \tan^2 A}{1 + \cot^2 A} =$
 (A) $\sec^2 A$ (B) -1 (C) $\cot^2 A$ (D) $\tan^2 A$
- Prove the following identities, where the angles involved are acute angles for which the expressions are defined.
 - $(\operatorname{cosec} \theta - \cot \theta)^2 = \frac{1 - \cos \theta}{1 + \cos \theta}$
 - $\frac{\cos A}{1 + \sin A} + \frac{1 + \sin A}{\cos A} = 2 \sec A$
 - $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \operatorname{cosec} \theta$
[Hint : Write the expression in terms of $\sin \theta$ and $\cos \theta$]
 - $\frac{1 + \sec A}{\sec A} = \frac{\sin^2 A}{1 - \cos A}$ **[Hint : Simplify LHS and RHS separately]**
 - $\frac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \operatorname{cosec} A + \cot A$, using the identity $\operatorname{cosec}^2 A = 1 + \cot^2 A$.
 - $\sqrt{\frac{1 + \sin A}{1 - \sin A}} = \sec A + \tan A$
 - $\frac{\sin \theta - 2 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta} = \tan \theta$
 - $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$